INCH-POUND

A-A-52092
March 14, 1991

COMMERCIAL ITEM DESCRIPTION

AMMONIUM CHLORIDE, TECHNICAL

The General Services Administration has authorized the use of this commercial item description in preference to O-A-491.

Abstract. This commercial item description covers technical ammonium chloride in brick form for use in cleaning soldering irons and in white crystal form for use in batteries and for galvanizing purposes.

1. CLASSIFICATION

1.1 Classification. This commercial item description covers ammonium chloride in the following types as specified (see 5.1).

Type I - Ammonium chloride bricks
Type II - Ammonium chloride white crystals

2. SALIENT CHARACTERISTICS

2.1 Physical and chemical requirements. The type I and type II ammonium chloride shall conform to the physical and chemical requirements specified in table I.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commander, U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-IRT, Natick, MA 01760-5019.

AMSC N/A FSC 6810

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TABLE I. Physical and chemical requirements

Requirement	Minimum	Maximum
Volatile matter (percent) 1/	---	1.0
Ammonium chloride content (percent) 2/	99.0	---
Residue on ignition (dry basis) (percent) 3/	---	0.8
Free acid 4/	Pass test	---
Heavy metals 5/	Pass test	---
Sieve particle size graduation (percent) 6/	90.0	---

1/ Place approximately 6 grams of the material, weighed to the nearest 0.1 milligram, into a flat-bottomed, tared dish, 2 to 3 inches in diameter, spreading the material over the bottom. Dry over sulfuric acid at room temperature (approximately 25°C) for 4 hours. Weigh to the nearest 0.1 milligram and calculate the percent loss as volatile matter as follows:

$$\text{Percent volatile matter} = \frac{100\ (W-R)}{W}$$

Where:

W = weight of specimen used, in grams
R = weight of residue after drying, in grams

2/ Dissolve approximately 0.2 grams of the dried specimen, weighed to the nearest 0.1 milligram, in approximately 40 milliliters of water. Add exactly 50 milliliters of 0.1 N silver nitrate solution and 5 milliliters of concentrated nitric acid. Make up the mixture to exactly 100 milliliters with water. Filter through a dry filter and reject the first 10 milliliters of the filtrate. Take exactly 50 milliliters of the filtrate and add 2 milliliters of an 8 percent solution of the ferric ammonium sulfate. Titrate the excess of silver nitrate in this portion with 0.1 N ammonium thiocyanate and calculate the percent ammonium chloride as follows:

$$\text{Percent ammonium chloride} = \frac{10.70\ (25N - VM)}{W}$$

Where:

N = normality of silver nitrate solution
V = volume of ammonium thiocyanate solution used, in milliliters
M = normality of ammonium thiocyanate solution
W = weight of specimen used, in grams

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3/ Weigh to the nearest 0.1 milligram approximately 5 grams of the dried specimen (see volatile matter) into a tared crucible. Add 2 milliliters of concentrated sulfuric acid. Heat at low temperature and then ignite gently to a constant weight. Calculate the percent residue as follows:

$$\text{Percent residue on ignition} = \frac{100\ R}{W}$$

Where:

R = weight of residue after ignition, in grams
W = weight of dried speciman used, in grams

4/ Weigh a 2 gram sample and dissolve in 20 milliliters of water adding one drop of methyl red indicator. Add one drop of 0.1 N sodium hydroxide. The yellow color shall remain.

5/ Dissolve 0.5 grams of the original specimen in 50 milliliters of water. Pass hydrogen sulfide gas through the solution and add 1 milliliter of ammonium hydroxide. No brown color shall develop. Disregard any green color present.

6/ Pass through a Number 30 sieve in accordance with ASTM E 11.

3. QUALITY ASSURANCE

3.1 Certification. The contractor shall certify, and maintain substantiating evidence, that the product offered meets the salient characteristics and requirements of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

3.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met.

4. PACKAGING

4.1 Packaging. Each type I, 8-ounce ammonium chloride brick shall be individually packaged in a polyethylene bag, heat sealed, and inserted into a snug-fitting, folding paperboard box. One pound of the type II ammonium chloride shall be packaged in a polyethylene bag, heat sealed, and inserted into a paperboard box.

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4.2 Packing.

4.2.1 Type I. Forty-eight pounds of the type I ammonium chloride; packaged as specified, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The shipping container shall be of a size palletizable in accordance with MIL-STD-147. Closure shall be in accordance with the appendix of the container specification.

4.2.2 Type II. Fifty pounds of the type II ammonium chloride shall be packed in a paper shipping sack conforming to type I, II, or IV, construction 4-4X of UU-S-48. Forty-eight 1-pound packages, packaged as specified, shall be packed in a fiberboard box conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636.

4.3 Palletization. When specified (see 5.1), ammonium chloride, packed as specified, shall be palletized in accordance with load type Ia or XV, as applicable, of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D, or film bonding means F or G.

4.4 Marking. In addition to any special markings required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

5. NOTES

5.1 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type required (see 1.1).
c. When palletization is required (see 4.3).

5.2 Sources of documents.

5.2.1 Source of government documents. Copies of Military and Federal documents are available from:

Standardization Documents Order Desk
Bldg. 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

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5.2.2 Source for non-Government document. Copies of ASTM E 11 are available from:

American Society for Testing and Materials (ASTM)
1916 Race Street
Philadelphia, PA 19103

MILITARY INTERESTS:

Custodians

Army - GL
Air Force - 68

Review activities

Army - AR, MD, SM
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CIVIL AGENCY COORDINATING ACTIVITIES:

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PREPARING ACTIVITY:

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